

Bachelor Thesis Self-Reflection Essay

My bachelor thesis set out to investigate whether Fisher-based gradient projection methods can be constructively integrated into chunked hypernetwork architectures for continual learning. The core tension I want to reflect on is this: a more thorough reading of the literature at the outset would have saved significant time, but it might also have prevented the discovery that became the thesis's primary novel contribution.

The clearest example of what insufficient upfront reading cost me came during hypernetwork experiments. FOPNG was consistently underperforming EWC by a wide margin, and I developed a hypothesis: FOPNG lacked any mechanism to anchor parameters that had been important for previous tasks. EWC penalises movement away from those parameters explicitly; FOPNG only constrains the gradient direction at update time. In a hypernetwork setting, I reasoned, this asymmetry would be particularly destructive because the same shared weights are reused across all tasks. This reasoning led me to design iFOPNG — replacing FOPNG's single-task Fisher metric with a combined metric that accumulates importance across tasks, implementing parameter inertia geometrically rather than through an explicit penalty.

That hypothesis was partially wrong. The more immediate reason FOPNG was underperforming was that I had not implemented von Oswald et al.'s hypernetwork regulariser, which I had treated as an optional add-on rather than a structural requirement. The regulariser penalises deviation of previously generated weights and is, in effect, an EWC-like consolidation mechanism native to the hypernetwork framework. Without it, the hypernetwork has no means of protecting earlier task representations at all, and any comparison between methods in that setting is measuring the absence of the regulariser more than the properties of the projection method. Once I understood this — late, and only because the performance gap had grown too large to explain otherwise — the experimental picture clarified considerably.

The uncomfortable reflection is that I should have read both papers carefully enough to understand this before running a single experiment. Had I done so, the experimental design would have been cleaner and weeks of debugging would have been unnecessary.

And yet I am not sure I would have conceived of iFOPNG through reading alone. The intuition that projection methods and parameter anchoring address different aspects of forgetting — and that combining them geometrically inside the Fisher metric is preferable to adding a separate penalty term — emerged from watching the methods fail in a specific way in a specific architecture. That kind of diagnostic observation tends to require contact with running systems, not just with papers.

The point of improvement I take from this is not simply "read more." It is more precise: read enough to correctly set up the experimental baseline before touching the code, but treat the exploratory phase as a legitimate source of hypotheses rather than as a failure of planning. The error was not that I experimented; it was that I had not done enough groundwork to correctly interpret what I was observing when things went wrong.